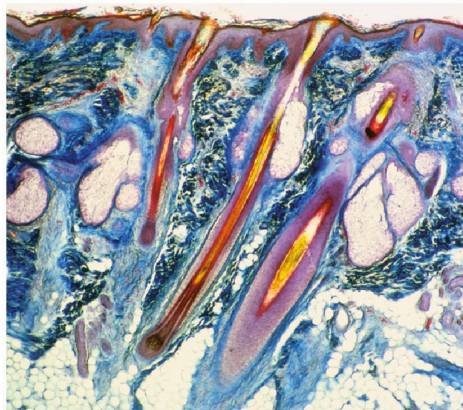


SKIN, HAIR, AND NAIL STRUCTURE

THE SKIN IS ONE OF THE LARGEST ORGANS OF THE BODY, WEIGHING 3–4KG (6–9LB) AND WITH A SURFACE AREA OF ALMOST 2M² (21SQ FT). IT IS FORMED FROM MANY TYPES OF CELL, SOME OF WHICH PRODUCE HAIR AND NAIL TISSUE.



SKIN SECTION

This micrograph shows three hair follicles and globules of sebum in the dermis (blue), with the thin epidermis (purple) on top.

SKIN STRUCTURE

The skin is not just a thin, waterproof covering but a complex organ consisting of a variety of specialized cells. The skin's thickness varies from about 0.5mm ($\frac{1}{50}$ in) on delicate areas such as the eyelids, to 5mm ($\frac{1}{5}$ in) or more on areas of wear and tear, such as the soles of the feet. Skin has two main layers: the outer epidermis, the main function of which is protection, and the underlying dermis. The dermis contains thousands of sensors that are sensitive to touch. It also contains sweat glands and blood vessels, which play a vital role in temperature regulation. Under the dermis is a layer known as subcutaneous fat. This acts as a buffer and provides insulation against extreme heat and cold.

SKIN RENEWAL

The epidermis continually renews and replaces itself. The basal layer consists of box-like cells that multiply quickly and are gradually pushed up to the surface by new cells forming below. As the cells move upwards, they develop tiny prickles that bind them together. They then flatten and fill with a waterproofing protein called keratin. Finally, the cells die, and reach the surface resembling interlocking roof tiles. As they flake away with wear and tear, more cells arrive to replace them.

EPIDERMAL LAYERS

The procession of skin cells from base to surface creates four layers (five in areas of great friction, such as the palms and soles) in the epidermis. As each cell moves upwards, it fills with keratin.

